Partial Algebraic Shifting

Antony Della Vecchia

Joint work with M. Joswig and F. Lenzen

TU Berlin

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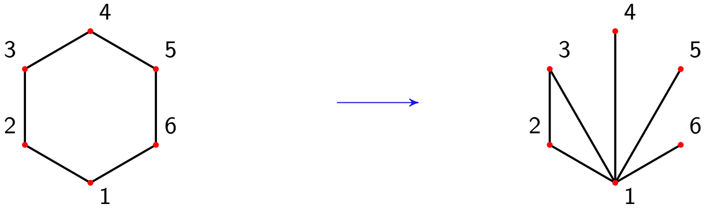
Algebraic Shifting

Definition
$C \in \binom{[n]}{k}$ is <u>shifted</u> if for any $\sigma \in C$, $\sigma|_i^j \in C$ for $i \in \sigma$ and $j < i$.



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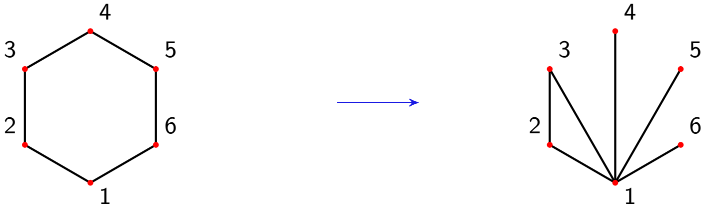

Theorem (Kalai '84, Björner-Kalai '88)

Given K a simplicial complex and a field $\mathbb{F}$, there exists a shifted simplicial complex $\Delta^{\mathbb{F}}(K)$ with the same f-vector and Betti numbers as K.

Exterior Algebraic Shifting

Definition

Let $\mathbb{E} \supset \mathbb{F}$ be a field extension and let $g \in \mathrm{GL}(n, \mathbb{E})$.

- The kth compound matrix $g^{\wedge k}$ is the $\binom{n}{k} \times \binom{n}{k}$-matrix of k-minors of g.
- $g^{\wedge K_k}$ is the $|K_k| \times \binom{n}{k}$-submatrix of $g^{\wedge k}$ with rows indexed by K_k.

Definition

The partial (exterior) shift of K by g is

$$\Delta_g^{\mathbb{F}}(K) := \bigcup_k \left\{ \sigma \in \binom{[n]}{k} : g_{*\sigma}^{\wedge K_k} \notin \mathrm{span}_{\mathbb{E}}\left((g_{*\rho}^{\wedge K_k})_{\rho <_{\mathrm{lex}} \sigma} \right) \right\}.$$

Small Example

Example

For $g = (x_{ij})_{1 \leq i,j \leq 4}$ and $K = \{\{1,2\},\{2,3\},\{4\}\}$, $\mathbb{F} = \mathrm{GF}(2)$.

$$g^{\wedge K_2} = \begin{pmatrix} x_{11}x_{22}+x_{12}x_{21} & x_{11}x_{23}+x_{13}x_{21} & x_{11}x_{24}+x_{14}x_{21} & x_{12}x_{23}+x_{22}x_{13} & x_{12}x_{24}+x_{14}x_{22} & x_{13}x_{24}+x_{14}x_{23} \\ x_{21}x_{32}+x_{22}x_{31} & x_{21}x_{33}+x_{23}x_{31} & x_{21}x_{34}+x_{24}x_{31} & x_{22}x_{33}+x_{23}x_{32} & x_{22}x_{34}+x_{24}x_{32} & x_{23}x_{34}+x_{24}x_{33} \end{pmatrix}.$$

Which row reduces to,

$$\begin{pmatrix} x_{11}x_{22}+x_{12}x_{21} & x_{11}x_{23}+x_{13}x_{21} & x_{11}x_{24}+x_{14}x_{21} & x_{12}x_{23}+x_{13}x_{22} & x_{12}x_{24}+x_{14}x_{22} & x_{13}x_{24}+x_{14}x_{23} \\ 0 & \begin{matrix} x_{11}x_{21}x_{22}x_{33} \\ +x_{11}x_{21}x_{23}x_{32} \\ +x_{12}x_{21}^2x_{33} \\ +x_{13}x_{21}^2x_{32} \\ +x_{12}x_{21}x_{23}x_{31} \\ +x_{13}x_{21}x_{22}x_{31} \end{matrix} & \begin{matrix} x_{11}x_{21}x_{22}x_{34} \\ +x_{11}x_{21}x_{24}x_{32} \\ +x_{12}x_{21}^2x_{34} \\ +x_{14}x_{21}^2x_{32} \\ +x_{12}x_{21}x_{24}x_{31} \\ +x_{14}x_{21}x_{22}x_{31} \end{matrix} & \begin{matrix} x_{11}x_{22}^2x_{33} \\ +x_{11}x_{22}x_{23}x_{32} \\ +x_{12}x_{21}x_{22}x_{33} \\ +x_{13}x_{21}x_{22}x_{32} \\ +x_{12}x_{22}x_{23}x_{31} \\ +x_{13}x_{22}^2x_{31} \end{matrix} & \begin{matrix} x_{11}x_{22}^2x_{34} \\ +x_{11}x_{22}x_{24}x_{32} \\ +x_{12}x_{21}x_{22}x_{34} \\ +x_{14}x_{21}x_{22}x_{32} \\ +x_{12}x_{22}x_{24}x_{31} \\ +x_{14}x_{22}^2x_{31} \end{matrix} & \begin{matrix} x_{11}x_{22}x_{23}x_{34} \\ +x_{11}x_{22}x_{24}x_{33} \\ +x_{12}x_{21}x_{23}x_{34} \\ +x_{12}x_{21}x_{24}x_{33} \\ +x_{13}x_{21}x_{24}x_{32} \\ +x_{14}x_{21}x_{23}x_{32} \\ +x_{13}x_{22}x_{24}x_{31} \\ +x_{14}x_{22}x_{23}x_{31} \end{matrix} \end{pmatrix}.$$

$\Delta(K) = \{\{1,2\},\{1,3\},\{4\}\}$

Generic Initial Ideals
For $\sigma = \{\sigma_1 < \ldots < \sigma_k\}$ set $e_\sigma = e_{\sigma_1} \wedge \ldots \wedge e_{\sigma_k}$

$$I_K := (e_\sigma : \sigma \notin K) \subset \bigwedge \mathbb{F}^n$$

The exterior face ring is,

$$\mathbb{F}[K] := \bigwedge \mathbb{F}^n \Big/ I_k,$$

and

$$\mathbb{F}[\Delta(K)] = \bigwedge \mathbb{F}^n \Big/ \mathrm{gin}(I_k)$$

See [Herzog-Tarai '99] Stable Properties of algebraic shifting.

Partial Algebraic Shifting

Consider now $\{\mathfrak{r}(w) \in \mathrm{GL}(n, \mathbb{E}) : w \in S_n\}$.
Let $c_n := (1 \dots n) \in S_n$.

Theorem (DV-Joswig-Lenzen '24+)

For any simplical complex K and $w \in S_n$ such that $c_n \leq w$ in the weak Bruhat order we have the following,

1. $\Delta_{\mathfrak{r}(w)}(K)$ *is homotopy equivalent to a wedge of spheres.*
2. $\Delta_{\mathfrak{r}(w)}(K)$ *has the same Betti numbers as* K*.*

Real Projective Plane

$K = \mathbb{RP}^2_6$, $\mathbb{F} = \mathbb{Q}$

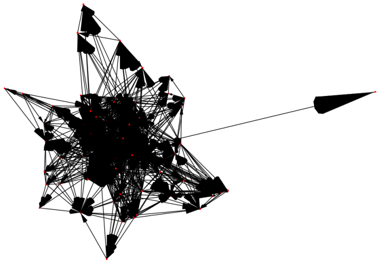

Figure: The partial shift graph starting from $\mathbb{RP}^2_6$, generated using `OSCAR`.

LU decomposition

Lemma

$\Delta_{gb}(K) = \Delta_g(K)$ for any upper triangular b and any simplicial complex K.

LU decomposition For $g \in \mathrm{GL}(n, \mathbb{E})$ with non zero leading principal minors, there exists l lower triangular, u upper triangular such that

$$g = lu$$

Example

$K = \{\{1,2\},\{2,3\},\{4\}\}$, and $I^{\wedge K_2}$ reduces to

$$\begin{pmatrix} x_{13}x_{24}+x_{14}x_{23} & x_{12}x_{24}+x_{14} & x_{24} & x_{12}x_{23}+x_{13} & x_{23} & 1 \\ 0 & \begin{matrix} x_{12}x_{23}x_{24}x_{34}+x_{12}x_{24}^2 \\ +x_{13}x_{24}x_{34}+x_{14}x_{24} \end{matrix} & \begin{matrix} x_{23}x_{24}x_{34} \\ +x_{24}^2 \end{matrix} & \begin{matrix} x_{12}x_{23}^2x_{34}+x_{12}x_{23}x_{24} \\ +x_{13}x_{23}x_{34}+x_{14}x_{23} \end{matrix} & \begin{matrix} x_{23}^2x_{34} \\ +x_{23}x_{24} \end{matrix} & \begin{matrix} x_{23}x_{34} \\ +x_{24} \end{matrix} \end{pmatrix},$$

Bruhat Decomposition

Bruhat Decomposition Let $B \subset \mathrm{GL}(n, \mathbb{E})$ be the subgroup of upper triangular matrices,

$$\mathrm{GL}(n, \mathbb{E}) \; = \coprod_{w \in S_n} BwB \; .$$

$$w_0 = (1 \;\; n)(2 \;\; n-1) \ldots (\lfloor \frac{n}{2} \rfloor \lceil \frac{n}{2} + 1 \rceil)$$

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If g is generic, then so is $w_0 g$, hence

$$\Delta_{w_0 g}(K) = \Delta_g(K) = \Delta(K).$$

For the same reason $w_0 g$ there exist some lower triangular l, and upper triangular u', u such that,

$$w_0 g = lu = w_0 u' w_0 u$$
$$g = u' w_0 u$$

$g \in Bw_0B$, and $\Delta_{u' w_0}(K) = \Delta(K)$

Inversions

Definition

For $w \in S_n$, let

$$\begin{aligned} \operatorname{Inv} w &:= \{(i,j) : i < j, i \cdot w > j \cdot w\} \\ \ell(w) &:= |\operatorname{Inv} w| \end{aligned}$$

$$w_0 = \operatorname{argmax}_{w \in S_n} \ell(w)$$

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Example

Take $w = (2\ \ 3)$, $\operatorname{Inv} w = \{(1,2),(1,3)\}$.

$$\begin{pmatrix} 1 & x & y \\ & 1 & z \\ & & 1 \end{pmatrix}\begin{pmatrix} 1 & & \\ & & 1 \\ & 1 & \end{pmatrix} = \begin{pmatrix} 1 & x & y \\ & 1 & z \\ & & 1 \end{pmatrix}\begin{pmatrix} 1 & & \\ & & 1 \\ & 1 & \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ & 1 & z \\ & & 1 \end{pmatrix}\begin{pmatrix} 1 & & \\ & & 1 \\ & 1 & \end{pmatrix}\begin{pmatrix} 1 & y & x \\ & 1 & 0 \\ & & 1 \end{pmatrix} .$$

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Minimal representatives

$$\mathfrak{u}(w) := \begin{pmatrix} 1 & u_{12} & \cdots & u_{1n} \\ & \ddots & \ddots & \vdots \\ & & 1 & u_{n-1,n} \\ & & & 1 \end{pmatrix} \quad \text{with} \quad u_{ij} = \begin{cases} x_{ij} & \text{if } (i,j) \in \operatorname{Inv} w, \\ 0 & \text{otherwise} \end{cases}.$$

We call $\mathfrak{r}(w) := \mathfrak{u}(w)w$ the <u>Rothe matrix</u>.

Weak Bruhat order

Definition (Björner-Brenti '05)

The weak Bruhat right order is a partial ordering on S_n,

$$w = uv \geq u$$

if $\ell(w) = \ell(u) + \ell(v)$.

Lemma

For $u, v \in S_n$ with $uv \geq u$ if and only if

$$\mathrm{Inv}\, uv = \mathrm{Inv}\, u \cup (\mathrm{Inv}\, v) \cdot u^{-1}$$
$$\emptyset = \mathrm{Inv}\, u \cap \mathrm{Inv}\, v$$

Rothe Matrix of the n cycle

Consider $c_n = (1\ \ 2 \ldots n)$,

- For $1 \leq i < j < n$, $i \cdot c_n < j \cdot c_n$
- $\operatorname{Inv} c_n = \{(i, n) : 1 \leq i < n\}$

$$\mathfrak{r}(c_n) = \begin{pmatrix} x_{1,n} & 1 & & \\ \vdots & & \ddots & \\ x_{n-1,n} & & & 1 \\ 1 & 0 & \cdots & 0 \end{pmatrix}.$$

For $w \geq c_n$

- $\operatorname{Inv} c_n \subset \operatorname{Inv} w$
- $\mathfrak{r}(c_n)$ and $\mathfrak{r}(w)$ have the same first column.

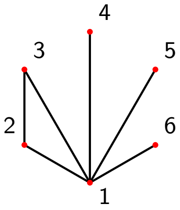
Near cones

Definition

A simplicial complex K is a near cone if for every $\sigma \in K$ with $1 \notin \sigma$ we have $\sigma|_i^1 \in K$ for each $i \in \sigma$.

Lemma (Björner-Kalai '88)

Let K be a near cone. Then K is homotopy equivalent to a wedge of spheres and $\beta_k(K) = |\{\sigma \in K^k : \sigma \cup \{1\} \notin K\}|$.



Permutation Lemma

Lemma (Björner-Kalai '88)

Let $g \in \mathrm{GL}(n, \mathbb{E})$ be generic and suppose for $\tau \in \binom{[n]}{k}$ there exists $\gamma \in \mathbb{E}$ such that,

$$g_\tau^{\wedge K_k} = \sum_{\sigma \in K_k} \gamma_\sigma g_\sigma^{\wedge K_k}.$$

If $\pi \in S_n$, then there exists $\gamma' \in \mathbb{E}$ such that

$$g_{\tau \cdot \pi}^{\wedge K_k} = \sum_{\sigma \in K_k} \gamma'_{\sigma \cdot \pi} g_{\sigma \cdot \pi}^{\wedge K_k}$$

Shifting to near cones

Lemma (DV-Joswig-Lenzen '24+)

Let K be a simplicial complex on n vertices. Further, let $g \in \mathrm{GL}(n, \mathbb{E})$ be a matrix whose entries in the first column are algebraically independent over $\mathbb{F}$ and the other entries of g. Then $\Delta_g(K)$ is a near cone.

Proof sketch

Let $\pi = (1 \;\; i)$ and suppose $\tau \notin K$,

$$g_{\tau}^{\wedge K_k} = \sum_{\sigma < \tau} \gamma_\sigma g_{\sigma}^{\wedge K_k}.$$

Then there exists $\gamma' \in \mathbb{E}$ such that,

$$g_{\tau \cdot \pi}^{\wedge K_k} = \sum_{\sigma < \tau \cdot \pi} \gamma'_\sigma g_{\sigma}^{\wedge K_k}$$

Preserving Betti Numbers

Lemma (DV-Joswig-Lenzen '24+)

Let $v, w \in S_n$ with $\ell(vw) = \ell(v) + \ell(w)$ then $\mathfrak{r}(vw)^{K_k}$ and $(\mathfrak{r}(v)\mathfrak{r}(w)^v)^{\wedge K_k}$. have the same column matroid.

Proof sketch of shifting by $w \geq c_n$ preserves Betti numbers
It is enough to show,

$$|\{\sigma \in \Delta_{\mathfrak{r}(w_0)}(K^k) : 1 \in \sigma\}| = |\{\sigma \in \Delta_{\mathfrak{r}(w)}(K^k) : 1 \in \sigma\}|.$$

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Use lemma and do some algebra in the subscripts of $\Delta_{\mathfrak{r}(w)}$ and $\Delta_{\mathfrak{r}(w_0)}$

Problem How and or when can we find $\Delta(K)$ from $\Delta_{\mathfrak{r}(c_n)}(K)$?

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`https://arxiv.org/abs/2410.24044`

Thank You!